Smartphone Usage Patterns and Their Association with Attention and Productivity among Young Adults

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Abstract

This study investigates the relationship between smartphone usage patterns and self-reported attention, productivity, and procrastination among individuals aged 18–30. With the increasing integration of smartphones into daily life, concerns have emerged regarding their potential impact on cognitive functioning and task-related behavior.

Data were collected through an anonymous online survey assessing daily screen time, social media use, phone behavior during study or work sessions, sleep duration, and stress levels. The sample included both students and young working adults, allowing for analysis of behavioral patterns across different contexts.

Statistical analyses, including Pearson correlation analysis, linear regression, mediation analysis, and k-means clustering, were conducted to examine associations between smartphone use and cognitive-behavioral outcomes.

Results showed no significant association between overall screen time and attention or productivity. However, specific smartphone usage patterns demonstrated weak associations with attention. Attention was positively associated with both productivity and procrastination, indicating a complex role in self-regulatory processes. Cluster analysis further revealed distinct behavioral profiles, with significant differences emerging only in procrastination.

Overall, the findings suggest that smartphone use is not uniformly associated with reduced cognitive performance, and that its relationship with attention and productivity depends on specific usage patterns and behavioral context.

1 Introduction

Smartphones have become an integral part of everyday life, particularly among young people. Originally designed as communication tools, they have evolved into multifunctional devices used for social interaction, entertainment, information retrieval, and academic activities. Their constant availability has raised growing interest in understanding how smartphone use may influence cognitive functioning and everyday productivity. Recent research indicates that smartphone use is not only frequent but deeply embedded in students' academic routines. Mobile devices are widely used during lectures, study sessions, and other learning-related activities. At the same time, modern applications increasingly incorporate attention-capturing design features such as notifications, recommendation algorithms, and visual cues, which encourage habitual checking behavior and compete for users' attentional resources [9]. As a result, smartphone interaction often occurs automatically rather than as a deliberate, goal-directed activity.

A growing body of literature has examined the relationship between smartphone use and academic and cognitive outcomes. Meta-analytic evidence suggests that problematic smartphone use is associated with lower academic performance, although the observed effects are generally small [1, 8]. One possible explanation for these mixed findings is that smartphone use is often treated as a single, uniform construct, despite substantial variation in how, when, and why individuals engage with their devices.

Indeed, recent evidence suggests that context-dependent patterns of smartphone use may be more strongly associated with negative outcomes than overall screen time. Behaviors such as smartphone multitasking during study sessions and checking messages while studying appear to show stronger relationships with lower academic performance than general usage measures [7]. These findings indicate that qualitative aspects of smartphone engagement may be more informative than simple measures of daily screen time.

Experimental studies further suggest that smartphones may influence cognitive functioning even in the absence of active use. It has been shown that the mere presence of a smartphone can reduce attentional control and increase cognitive distraction, suggesting that smartphones may impose cognitive costs through attentional capture mechanisms [6]. Such findings highlight the importance of examining attention as a potential mechanism linking smartphone-related behavior to broader behavioral outcomes.

Despite accumulating evidence, several gaps remain in the literature. Most studies focus either on general usage metrics or academic performance outcomes, while less attention has been given to specific behavioral usage patterns and their relationship with cognitive mechanisms. In addition, it remains unclear whether context-specific forms of smartphone use explain variability in outcomes more effectively than global indicators such as screen time.

The present study does not aim to determine whether smartphones are universally harmful. Instead, it investigates whether specific patterns of smartphone use—including daily screen time, phone use during studying, morning phone checking habits, and perceived difficulty disengaging from the device—are associated with self-reported attention and productivity among students. Furthermore, the study examines whether attention mediates the relationship between smartphone use patterns and productivity, and whether distinct behavioral profiles of smartphone users can be identified.

The study addresses the following research questions:

RQ1: Are global measures (screen time) stronger predictors of attention and productivity than context-dependent measures (phone use during studying, morning phone checking)?

RQ2: Does attention mediate the relationship between smartphone use patterns and productivity?

RQ3: Can distinct behavioral profiles of smartphone users be identified that differ in attention and productivity?

2 Related Works

2.1 Smartphone usage and academic performance

Empirical research on the relationship between smartphone use and academic outcomes has accumulated a substantial body of evidence; however, findings remain mixed. Meta-analytic studies provide the most comprehensive overview. For example, a meta-analysis of 29 studies including approximately 49,000 students reported a small but statistically significant negative association between problematic smartphone use and academic performance ($r = 0.110$, $p < .001$). Notably, this effect appears to be more pronounced in primary and secondary school students, suggesting that age may be an important moderating factor [8].

These findings are consistent with an earlier systematic review by Amez and Baert [1], which reported that 18 out of 23 studies (78.3%) found a negative association between smartphone use and academic performance. However, the authors also highlighted an important methodological limitation: studies relying on self-reported GPA were more likely to report null findings compared to those using objective academic records.This suggests that measurement approaches for both independent (smartphone use) and dependent (academic performance) variables may systematically influence observed results.

Further nuance is provided by Sunday et al. [7], who meta-analyzed 44 studies and demonstrated that not all types of smartphone use are equally detrimental. The strongest negative associations were observed for multitasking with smartphones during academic tasks ($r = -0.16$) and texting during class ($r = -0.13$), whereas general screen time showed a weaker association ($r = -0.10$). These findings suggest that how and when smartphones are used may be more important than overall usage duration.

2.2 Attention and cognitive impact

While Section 2.1 focuses on academic outcomes as behavioral endpoints, a logical next step is to examine the cognitive mechanisms through which smartphone use may exert its effects. Attention plays a central role in this process. Experimental and behavioral evidence suggests that smartphone use may influence attentional control. However, findings are not fully consistent. Hartanto et al. [4], comparing screen time ($\beta = -0.07$, $p = .381$) and checking frequency ($\beta = 0.03$, $p = .730$), did not find significant associations with inhibitory control.These inconsistencies may be explained by methodological differences: experimental studies often capture short-term distraction effects, whereas observational data reflect habitual usage patterns that may be moderated by individual differences in self-regulation.

At the same time, laboratory studies indicate that cognitive effects of smartphones may occur even in the absence of active use. Schwaiger and Tahir [6] found that participants performed significantly better on an inhibitory control task (Stroop test) when their smartphones were placed in another room compared to conditions where the device was on the desk or in a pocket ($F(3, 134) = 3.128$, $p < .05$).This "mere presence" effect suggests that cognitive costs may arise not only from direct interaction with smartphones but also from attentional allocation toward the device.

Further evidence highlights habitual and context-dependent mechanisms of distraction. Oulasvirta et al. [5] demonstrated that smartphone checking behavior often occurs during low cognitive load situations, such as waiting, boredom, or passive engagement (e.g., lectures), forming habitual interruption patterns that disrupt sustained attention.

These effects may be reinforced by design features of digital environments. Chen et al.[9] showed that persuasive design elements such as notifications, visual cues, and algorithmic recommendations encourage automatic checking behavior every 15-20 minutes, thereby fragmenting attention and reinforcing habitual engagement cycles. At a broader cognitive level, Busch and McCarthy[16], in a systematic review, reported that problematic smartphone use is associated with inattentional blindness and reduced situational awareness, indicating more general disruptions in attentional allocation and stability. Finally, Almourad et al.[13], in an analysis of 47 definitions of digital addiction, identified a cluster of attention-related symptoms—including loss of control, intrusive thoughts about digital content, distorted time perception, and compulsive checking—as core diagnostic features, further emphasizing the central role of attentional disruption in smartphone-related behaviors.

2.3 Productivity and behavioral outcomes

If attention is considered a key cognitive mechanism through which smartphone use affects behavior, then productivity and procrastination represent key downstream outcomes reflecting task execution and self-regulation. Procrastination is traditionally conceptualized as a failure of self-regulation. Steel [21] defines procrastination as the irrational delay of intended tasks despite expecting negative consequences, emphasizing deficits in self-regulatory control as a central mechanism. However, procrastination can also be understood as an emotion regulation strategy, particularly in response to tasks perceived as stressful, cognitively demanding, or aversive. Under such conditions, individuals may avoid task engagement to reduce negative affect and redirect attention toward more immediately rewarding and less demanding stimuli, including smartphone use.

Empirical evidence supports this perspective. Esan et al. [2] found that problematic smartphone use is associated with lower academic performance, particularly under multitasking conditions.Their study reported that 51.6% of students occasionally lost track of time while using smartphones, and 21.7% delayed academic tasks due to phone use, reflecting direct manifestations of procrastination and impaired self-regulation. Similarly, Chen et al. [9] reported that 44% of participants perceived smartphones as negatively affecting their academic or professional performance.

At a broader level, Busch and McCarthy[16] concluded in their systematic review that problematic smartphone use is consistently associated with reduced productivity through procrastination, impaired time management, and multitasking, all of which undermine sustained goal-directed behavior. Taken together, these findings suggest a cognitive-executive cascade in which attentional disruption contributes to reduced cognitive control, ultimately resulting in decreased self-regulation, increased procrastination, and lower productivity.

2.4 Research gap: contradictions and open questions

Despite growing evidence, the literature remains fragmented and contradictory. Experimental studies such as Schwaiger and Tahir [6] demonstrate that even the mere presence of a smartphone can impair inhibitory control. However, Hartanto et al. [4] found no significant relationship between objectively measured screen time ($\beta = -0.07$, $p = .381$)

or checking frequency ($\beta = 0.03$, $p = .730$) and inhibitory control. These divergent findings may reflect methodological differences: experimental paradigms capture potential distraction effects, whereas observational measures reflect actual usage patterns that are influenced by individual differences in self-regulation.

Supporting this interpretation, Sarker et al. (2025) found that 69.7% of students believe they overuse their smartphones, while only 20.2% report using time-management applications, suggesting limited self-awareness and discrepancies between perceived and actual usage.

Furthermore, construct definition plays an important role. For instance, introducing a construct such as "pervasiveness"—defined as the frequency of smartphone use across academic, sleep, and social contexts—may better predict academic performance than subjective addiction scales. This suggests that global indicators such as "screen time" or "smartphone addiction" may be context-dependent and less closely aligned with behavioral outcomes than specific usage patterns [3].

Importantly, most studies examine cognitive and behavioral outcomes in isolation. Research on attentional effects [6], [4] rarely includes direct measures of productivity, while studies on academic performance [1], [8] often lack cognitive variables such as attention. As a result, the potential mediating role of attention - whether smartphone use affects productivity through attentional disruption - remains largely underexplored.

Additionally, user populations are often treated as homogeneous. Studies by Oulasvirta et al. [5] and Noë et al. [15] have highlighted meaningful differences between types of smartphone interaction (e.g., passive scrolling vs. active engagement), suggesting that aggregating all usage into a single construct may obscure important behavioral variation. Finally, Busch and McCarthy [16] note that existing models of problematic smartphone use explain only 10-27% of variance in cognitive and behavioral outcomes. This indicates substantial unexplained variability, likely driven by individual usage patterns and contextual factors that are not adequately captured in most studies.

2.5 Hypotheses

The literature suggests that smartphone usage may be associated with both cognitive performance and behavioral outcomes such as attention and productivity. However, findings remain mixed, and recent studies emphasize that not only the amount of smartphone use, but also usage patterns and psychological dependence may play an important role. In particular, factors such as habitual phone checking and perceived discomfort when being without a smartphone may further explain individual differences in attention and productivity outcomes among young adults. Based on these findings, the following hypotheses are proposed:

H1: Smartphone usage is negatively associated with attentional performance.

H2: Attentional performance is positively associated with productivity and negatively associated with procrastination.

H3: Attentional performance mediates the relationship between smartphone usage and productivity-related outcomes.

3 Methodology

3.1 Participants

Participation in the study was voluntary and anonymous. The online questionnaire was distributed through social media platforms and online communities and targeted individuals aged 18-30. Both students and non-students were eligible to participate.

A total of 109 responses were collected. After data cleaning, 15 responses were excluded due to incomplete questionnaires ($n = 8$) and age outside the target range ($n = 7$). The majority of respondents were students. In terms of gender distribution, the sample included both male and female participants, with a slight predominance of female respondents. Most participants were concentrated in the younger age range (18-24 years), while a smaller proportion belonged to the 25-30 age group. Detailed sample characteristics are reported in Table 1.

Table 1: Demographics

Variable	***M* (SD)**	**Frequency (%)**
Age	22.09 (2.46)	
Sex		
Male		42 (44.7%)
Female		52 (55.3%)
Education Level		
Bachelor's student		57 (60.6%)
Master's student		19 (20.2%)
Employed		18 (19.1%)
Daily screen time (hours)	6.94 (4.10)	
Sleep duration (hours)	6.85 (0.91)	

3.2 Measurement Tools

The survey included self-reported measures of smartphone and social media usage, attention, productivity, and lifestyle factors. Smartphone usage was assessed using questions on average daily screen time and frequency of phone use during study or work sessions.

Smartphone use was assessed using three indicators: average daily screen time (1 = "Less than 2 hours" to 5 = "More than 5 hours"), frequency of phone use during study or work sessions (1 = "Never" to 5 = "Always"), and a binary indicator of whether participants checked their phone within five minutes of waking (0 = "No", 1 = "Yes").

Attention and productivity were measured using self-report Likert-scale items, ranging from 1 ("Strongly disagree") to 5 ("Strongly agree"). The attention composite was calculated as the mean of three items: sustained concentration ability, loss of concentration due to smartphone use (reverse-coded), and difficulty disengaging from the smartphone during task performance (reverse-coded). The productivity composite was calculated as the mean of two items: perceived productivity during study or work sessions and procrastination (reverse-coded).

In the final sample, the attention composite demonstrated acceptable internal consistency (Cronbach's $\alpha = .72$), while the productivity composite showed moderate reliability (Cronbach's $\alpha = .68$).

3.3 Data Preprocessing

The initial dataset included 109 responses. All data were exported to .csv format and processed in Python using the pandas and numpy libraries. During data cleaning, 15 responses were excluded: 8 due to missing values and 7 due to age outside the inclusion criterion (18–30 years). The final analytical sample consisted of 94 participants.

Variables reflecting smartphone usage frequency were converted from categorical responses into ordinal numerical scales ranging from 1 to 5. The "morning phone checking" variable was coded as binary (1 = Yes, 0 = No).

Based on these recoded variables, composite indices for attention and productivity were constructed as described above.

3.4 Statistical Analysis

All statistical analyses were conducted using JASP (version 0.18) and Python. Prior to inferential analyses, assumptions of normality, linearity, homoscedasticity, and multicollinearity were assessed.

Descriptive statistics. Means (M), standard deviations (SD), frequencies, and percentages were calculated for all study variables. Correlation analysis. To test H1 ("Smartphone use is negatively associated with attention") and H2 ("Attention is positively associated with productivity and negatively associated with procrastination"), Pearson correlation coefficients (r) were computed. A correlation matrix was generated for all numerical variables. Statistical significance was set at ($p < .05$).

Multiple linear regression. To address RQ1 ("Are global smartphone use metrics stronger predictors than context-dependent metrics?"), multiple linear regression analysis was conducted with productivity as the dependent variable. Predictor variables included daily screen time, phone use during study/work sessions, morning phone checking, attention, and stress level. For the regression model, R^2, standardized regression coefficients β, and p-values were reported.

Mediation analysis. To test H3 ("Attention mediates the relationship between smartphone use and productivity"), mediation analysis was conducted using bootstrapping (5,000 resamples) to estimate indirect effects. Smartphone use served as the independent variable (X), attention as the mediator (M), and productivity as the dependent variable (Y). Mediation was considered significant if the 95% confidence interval for the indirect effect did not include zero.

Cluster analysis. To identify behavioral user profiles (RQ3), K-means clustering was applied. The optimal number of clusters was determined using the elbow method, supplemented by silhouette and Davies-Bouldin indices. Variables included in the clustering procedure were daily screen time, phone use during study/work, morning phone checking, and stress level. After cluster identification, group differences in attention and productivity were examined using one-way ANOVA.

4 Results

4.1 Descriptive statistics

Descriptive statistics for all key variables are presented in Table 2. The mean daily screen time was 3.42 (SD = 0.90) on a 5-point scale, corresponding to approximately 4–6 hours per day. Social media use (M = 2.78, SD = 0.87), sleep duration (M = 3.18, SD = 0.70), and phone use during studying (M = 3.20, SD = 0.88) were at moderate levels. Morning phone checking was reported by 85.1% of respondents. The average stress level was 2.22 (SD = 1.13) on a scale from 0 to 5.

The composite attention score had a mean of 2.71 (SD = 0.72), with acceptable internal consistency (Cronbach's $\alpha = .72$). Productivity (M = 3.33, SD = 0.97) and procrastination (M = 2.09, SD = 1.27) were analyzed as single-item measures because the two-item productivity scale demonstrated poor internal consistency (inter-item $r = .091$).

Table 2: Descriptive Statistics of Key Variables

Variable	***M***	***SD***	**Min**	**Max**	α
Smartphone Usage					
Screen Time	3.42	0.90	2	5	
Social Media	2.78	0.87	1	5	
Usage During Study/Work	3.20	0.88	1	5	
Morning Check	0.85	0.36	0	1	
Lifestyle Factors					
Sleep Duration	3.18	0.70	2	5	
Stress Level	2.22	1.13	2	5	
Cognitive and Behavioral Outcomes					
Attention	2.71	0.72	1.33	4.67	.72
Productive	3.33	0.97	1	5	
Procrastination	2.09	1.27	1	5	

Note. Smartphone usage, sleep duration, and stress level were measured on 1–5 scales. Morning Check was coded as 0 = No, 1 = Yes. Attention is a composite score (Cronbach's $\alpha = .72$). Productivity and Procrastination are single-item indicators.

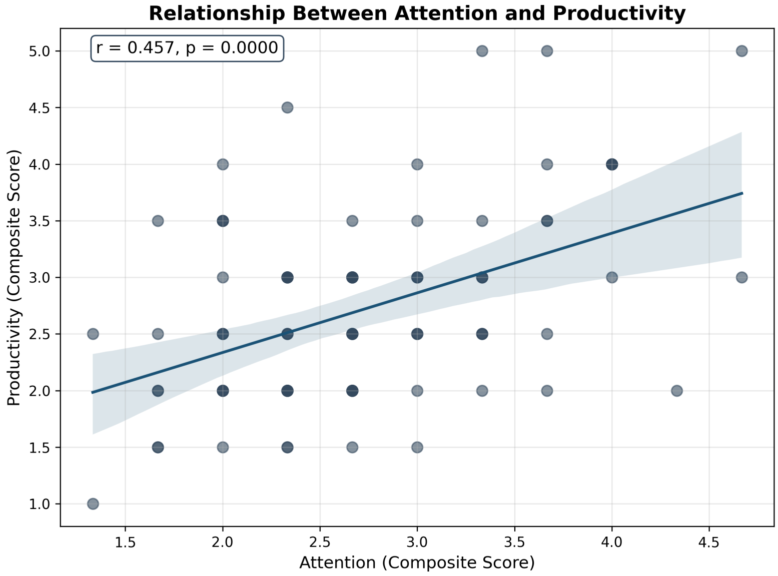
4.2 Correlation analysis

Pearson correlation analysis was conducted to examine relationships between the study variables (Table 3). The strongest positive correlation was observed between attention and productivity ($r = 0.457$, $p < .001$). Moderate positive correlations were also found between screen time and social media use ($r = 0.465$, $p < .001$), stress and phone use during studying ($r = 0.357$, $p < .001$), and stress and attention ($r = 0.298$, $p < .01$).

Table 3: Pearson's Correlation Matrix with Significance

	ST	SM	SU	MC	SL	AT	PR
ST	**1.000*** **	0.465***	0.221*	0.161	0.333**	0.014	0.006
SM	0.465***	**1.000*** **	0.357***	0.203*	0.293**	0.295**	0.206*
SU	0.221*	0.357***	**1.000*** **	0.097	0.357***	0.204*	0.245*
MC	0.161	0.203*	0.097	**1.000*** **	0.137	0.218*	0.123
SL	0.333**	0.293**	0.357***	0.137	**1.000*** **	0.298**	0.323**
AT	0.014	0.295**	0.204*	0.218*	0.298**	**1.000*** **	0.457***
PR	0.006	0.206*	0.245*	0.123	0.323**	0.457***	**1.000*** **

Note. ST = Screen Time, SM = Social Media, SU = Study Usage, MC = Morning Check, SL = Stress Level, AT = Attention, PR = Productivity.
$^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.



To visualize the main relationship between attention and productivity, a scatter plot with linear regression line was constructed (Figure 1). The plot illustrates a positive association between the two variables, indicating that higher attention scores were associated with higher self-reported productivity.

4.3 Hypothesis Testing

H1 predicted a negative relationship between smartphone usage and attention. This hypothesis was not supported. Total screen time was not significantly associated with attention ($r = 0.014$, $p = .897$). However, several specific smartphone usage behaviors showed weak but statistically significant positive associations with attention, including social media use ($r = 0.295$, $p = .004$), smartphone use during study ($r = 0.204$, $p = .049$), and morning phone checking ($r = 0.218$, $p = .035$).

H2 proposed that attention would be positively associated with productivity and negatively associated with procrastination. This hypothesis was partially supported. Attention was significantly positively correlated with productivity ($r = 0.208$, $p = .044$), consistent with the expected direction. However, attention was also significantly positively associated with procrastination ($r = 0.440$, $p < .001$), contrary to the hypothesis.

H3 H3 proposed that attention mediates the relationship between smartphone usage and productivity. This hypothesis was not supported. The prerequisite relationships were not observed, as smartphone usage showed no meaningful association with attention (c = 0.029; a = 0.011). Although attention was significantly associated with productivity (b = 0.279), the indirect effect was small (ab = 0.004), and the 95% bootstrap confidence interval included zero [-0.054, 0.070], indicating no significant mediation effect.

4.4 Cluster analysis

K-means clustering was conducted to identify behavioral user profiles. The optimal number of clusters was determined using the elbow method, silhouette coefficient, and Davies-Bouldin index. As shown in Table 4, the silhouette coefficient reached its highest value at $k = 3$ (0.376), while Davies-Bouldin index reached its lowest value (1.033). Based on these metrics, a three-cluster solution was selected for further analysis. Mean values for each cluster are presented in Table 5.

Table 4: Comparison of Cluster Metrics

Cluster (k)	Silhouette Score	Davies-Bouldin Index
2	0.306	1.348
3	0.376	1.033
4	0.305	1.138
5	0.308	0.956
6	0.333	1.065

Cluster 0 ($n = 14$) was characterized by moderate screen time (3.07), moderate phone use during studying (3.00), no morning phone checking, relatively low stress (1.86), and the lowest attention (2.33) and productivity (2.46) scores among all clusters.

Cluster 1 ($n = 28$) demonstrated the highest levels of screen time (4.29), phone use during studying (3.86), stress (3.36), attention (2.91), and productivity (2.98). All participants in this cluster reported checking their phones in the morning.

Cluster 2 ($n = 52$) the largest group, showed relatively low screen time (3.04), the lowest phone use during studying (2.90), and the lowest stress level (1.71). Similar to Cluster 1, all participants in this group reported morning phone checking.

Table 5: Cluster Profiles

Variable	Cluster 0 ($n = 14$)	Cluster 1 ($n = 28$)	Cluster 2 ($n = 52$)
Screen Time	3.07	4.29	3.04
Study Usage	3.00	3.86	2.90
Morning Check	0.00	1.00	1.00
Stress	1.86	3.36	1.71
Attention	2.33	2.91	2.70
Productivity	2.46	2.98	2.63

The majority of respondents (55.3%) belonged to Cluster 2, while Cluster 1 accounted for 29.8% and Cluster 0 represented the smallest group (14.9%).

A one-way ANOVA was conducted to examine differences between clusters in attention, productivity, and procrastination. Results are presented in Table 6. No statistically significant differences were found between clusters in attention, $F(2, 91) = 3.08$, $p = .051$, although the result approached statistical significance. Similarly, no significant differences were observed for productivity, $F(2, 91) = 0.23$, $p = .794$.

For attention, no statistically significant differences were found between clusters, $F(2, 91) = 3.078$, $p = .051$. Similarly, no significant differences were observed for productivity, $F(2, 91) = 0.231$, $p = .794$, indicating that cluster membership did not meaningfully differentiate participants in terms of attentional capacity or self-reported productivity.

In contrast, a significant effect of cluster membership was found for procrastination, $F(2, 91) = 3.97$, $p = .022$, with a moderate effect size ($\eta^2 = .080$). Post-hoc Tukey HSD comparisons indicated that Cluster 1 ($M = 2.57$) reported significantly higher levels of procrastination compared to Cluster 0 ($M = 1.50$, $p = .025$), while no other pairwise differences were statistically significant.

These findings suggest that cognitive outcomes such as attention and productivity remain relatively stable across behavioral profiles, whereas differences emerge primarily in behavioral self-regulation, particularly procrastination.

Table 6: ANOVA and Post-hoc Tests

Variable	Cluster 0	Cluster 1	Cluster 2	$F(2, 91)$	p	η^2
Attention	2.33	2.91	2.70	3.08	.051	.063
Productivity	2.46	2.98	2.63	0.23	.794	.005
Procrastination	1.50	2.57	1.98	3.97	.022	.080

5 Discussion

This study investigated the relationship between smartphone use, attention, productivity, and procrastination, as well as potential behavioral profiles of smartphone users among young adults. Overall, the findings provide a nuanced picture in which smartphone use is not uniformly associated with cognitive and behavioral outcome, and where attention plays a more complex role than initially hypothesized.

5.1 Smartphone use and attention

H1 proposed a negative association between smartphone use and attention. This hypothesis was not supported by the results. Overall screen time was not significantly related to attention, suggesting that general smartphone exposure does not necessarily impair attention capacity in this sample. However, specific usage usage patterns showed weak but statistically significant positive associations with attention, including social media use, phone use during studying, and morning phone checking.

These findings suggest that the relationship between smartphone use and attention may be more complex than a simple linear association with total screen time. The absence of a negative relationship indicates that overall smartphone exposure alone is not sufficient to explain variations in attention in this sample.

5.2 Attention, productivity, and procrastination

H2 proposed that attention would be positively associated with productivity and negatively associated with procrastination. This hypothesis was partially supported. Attention was positively associated with productivity, consistent with expectations, indicating that higher attentional capacity is related to better perceived task performance.

However, attention was also positively associated with procrastination, which contradicts the initial hypothesis. Rather than indicating a measurement error, this finding may reflect a more complex role of attention in self-regulatory processes. Based on prior theoretical work by Piers Steel [21], procrastination can be understood not only as a failure of self-regulation but also as an emotion regulation strategy. From this perspective, individuals with higher attentional awareness may also experience greater task salience, cognitive load, or performance pressure, which can increase avoidance behaviors despite maintained cognitive engagement.

Thus, attention in this study appears to function as a dual-factor construct, being associated with both adaptive (productivity) and maladaptive (procrastination) outcomes.

5.3 Mediation hypothesis

H3 proposed that attention would mediate the relationship between smartphone use and productivity. This hypothesis was not supported. Smartphone use was not significantly associated with either attention or productivity, which prevented the establishment of a mediation pathway. Although attention was positively associated with productivity, the indirect effect of smartphone use on productivity via attention was not statistically significant.

These results suggest that attention does not function as a mediating mechanism between smartphone use and productivity in this sample. Instead, productivity appears to be more directly related to attentional capacity rather than to smartphone use patterns.

5.4 Behavioral profiles

Cluster analysis identified three distinct behavioral profiles. However, cluster membership did not significantly differentiate participants in terms of attention or productivity. Both variables remained relatively stable across clusters, with no statistically significant differences observed, although attention showed a marginal effect.

In contrast, cluster membership was significantly associated with procrastination. Participants in the high-use, high-stress cluster reported significantly higher levels of procrastination compared to the low-use cluster. No other pairwise differences were significant. This suggests that behavioral differences between smartphone user profiles are primarily reflected in self-regulatory outcomes rather than cognitive outcomes.

5.5 Limitations

Several limitations should be considered when interpreting the findings of this study.

First, the cross-sectional design does not allow for causal inferences. Although associations between smartphone use, attention, productivity, and procrastination were identified, it is not possible to determine the directionality of these relationships.

Second, the study relied primarily on self-report measures. This may introduce response biases, such as social desirability or subjective over- or underestimation of smartphone use and behavioral outcomes. In particular, constructs such as attention, productivity, and procrastination may be influenced by individual perception rather than objective performance indicators.

Third, some variables were measured using brief or single-item scales. While this approach allowed for efficient data collection, it may have limited the reliability and precision of measurement, particularly for productivity and certain behavioral indicators.

Fourth, the sample size within some clusters was relatively small, especially Cluster 0 ($n = 14$). This may reduce statistical power for detecting between-group differences and limits the generalizability of cluster-specific findings.

Finally, the sample consisted of university students, which may limit the generalizability of the results to other populations. Smartphone use patterns and self regulatory behaviors may differ across age groups, educational contexts, and cultural settings.

6 Conclusion

Overall, the findings suggest that smartphone use does not exert a uniform or directly negative influence on attention or productivity. Instead, cognitive and behavioral outcomes appear to depend on more specific patterns of use and individual differences in attentional functioning.

Importantly, attention emerged as a central variable in the model, being positively associated with both productivity and procrastination. This pattern indicates that attention may not operate solely as a protective cognitive resource but may also co-occur with increased task awareness and psychological pressure that contribute to procrastination.

Taken together, the results support a more differentiated view of smartphone-related behaviors, in which effects are not uniformly detrimental but vary depending on behavioral context and individual cognitive characteristics.

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